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The validity and reliability of the Indonesian version of the Godin-Shepard's Leisure time physical activity questionnaire

Farid Rahman[1ABCE], Brissa Isabel Salsabila[1BCD], Ilham Setya Budi[2ABE], Dessy Triana Ramadhani[1BCD], Syarif Hidayat[1BCD], Azliyana Azizan[3ABC]

[1] Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta, Indonesia
[2] Dr. Muh. Sani Hospital, Kepulauan Riau, Indonesia
[3] Department of Physiotherapy, Faculty of Health Science, Universiti Teknologi Mara Shah Alam, Malaysia

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Correspondent author: Farid Rahman, fr280@ums.ac.id, https://orcid.org/0000-0001-7397-4306, Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta, Indonesia

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Abstract

Background and purpose

The World Health Organization (2020) suggests 150 minutes of moderate-level aerobic physical activity per week or 75 minutes of solid intensity per week. Every physical activity must be assessed with measuring equipment to determine how active a person is in their spare time. However, the Questionnaire instrument is only available in a few languages and has yet to be in Indonesian. Furthermore, similar questionnaires have yet to be translated into Indonesian, focusing on measuring exercise or sports activity. In addition, some of the physical activity or physical exercise content listed in the questionnaire differs from Indonesia's geographic and cultural conditions, such as cross-country skiing, judo, roller skating, alpine skiing, and snow-mobilizing.

This study aimed to assess young people's physical activity levels. In light of this description, the researcher wishes to investigate the reliability and validity of the Godin-Shephard Leisure-Time Physical Activity Questionnaire

Material and methods

The respondents who participated in this research were 74 young adults aged 16-25. Construct validity uses item analysis, convergent validity was determined through correlation with the Global Physical Activity Questionnaire. The test-retest analysis uses correlation on respondents who repeat filling in the questionnaire, and inter-rater reliability is determined through the consistency of three raters who have been determined with specific criteria.

Results

Pearson's correlation and the Intraclass Correlation Coefficient (ICC) were used to analyze the data. The construct validity showed r> 0.5, and the retest reliability was 0.998, indicating that the questionnaire was valid and reliable, and the ICC was 0.943, indicating that the Godin-Shepard's Leisure Time Physical Activity Questionnaire had excellent interrater consistency.

Conclusions

According to the research, the Godin-Shepard's Leisure Time Physical Activity Questionnaire conditions Indonesian is valid and trustworthy for use in young adult individuals.

Key words: physical activity questionnaire, validity, reliability, young adult, Indonesian Adaptation

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Анотація

Фарід Рахман, Брісса Ізабель Салсабіла, Ільхам Сетья Буді, Дессі Тріана Рамадані, Сяріф Хідаят, Азляна Азізан. Обґрунтованість і надійність індонезійської версії опитувальника фізичної активності у вільний час Годіна-Шепарда

Обгрунтування і мета

Всесвітня організація охорони здоров'я (2020) пропонує 150 хвилин аеробної фізичної активності помірного рівня на тиждень або 75 хвилин інтенсивного фізичного навантаження на тиждень. Будь-яка фізична активність повинна оцінюватися за допомогою вимірювального обладнання, щоб визначити, наскільки активна людина у вільний час. Однак анкета доступна лише кількома мовами, а індонезійською мовою її ще немає. Крім того, індонезійською мовою ще не перекладено аналогічні опитувальники, орієнтовані на вимірювання фізичної або спортивної активності. Крім того, деякі види фізичної активності або фізичних вправ, перелічені в анкеті, відрізняються від географічних та культурних умов Індонезії, наприклад, бігові лижі, дзюдо, катання на роликових ковзанах, гірські лижі та снігохідний спорт.

Метою цього дослідження була оцінка рівня фізичної активності молоді. У світлі цього опису дослідник хотів би вивчити надійність та валідність опитувальника Годіна-Шепхарда "Фізична активність у вільний час".

Матеріал та методи

У дослідженні взяли участь 74 молоді дорослі особи віком 16-25 років. Для визначення конструктної валідності використовувався аналіз елементів, конвергентна валідність визначалася через кореляцію з Глобальним опитувальником фізичної активності. Для аналізу " тест-ретест " використовувалася кореляція з респондентами, повторно заповнюють опитувальник, а межретестовая надійність визначалася через узгодженість дій трьох рейтерів, визначених з допомогою спеціальних критеріїв.

Результати

Для аналізу даних використовувалися кореляція Пірсона та коефіцієнт внутрішньокласової кореляції (ICC). Конструктивна валідність показала r>0,5, ретестова надійність - 0,998, що свідчить про валідність та надійність опитувальника, а ICC - 0,943, що вказує на відмінну міжретестову узгодженість опитувальника Годіна-Шепарда "Фізична активність у вільний час".

Висновки

Згідно з результатами дослідження, опитувальник Годіна-Шепарда "Фізична активність у вільний час" індонезійською є валідним і надійним для використання в осіб молодого віку.

Ключові слова: опитувальник фізичної активності, валідність, надійність, молоді люди, індонезійська адаптація

Аннотация

Фарид Рахман, Брисса Исабель Салсабила, Ильхам Сетья Буди, Десси Триана Рамадхани, Сиариф Хидаят, Азлияна Азизан. Валидность и надежность индонезийской версии опросника Година-Шепарда по физической активности в свободное время

Обоснование и цель

Всемирная организация здравоохранения (2020) предлагает 150 минут аэробной физической активности умеренного уровня в неделю или 75 минут интенсивной физической нагрузки в неделю. Любая физическая активность должна оцениваться с помощью измерительного оборудования, чтобы определить, насколько активен человек в свободное время. Однако анкета доступна только на нескольких языках, а на индонезийском языке ее еще нет. Кроме того, на индонезийский язык еще не переведены аналогичные опросники, ориентированные на измерение физической или спортивной активности. Кроме того, некоторые виды физической активности или физических упражнений, перечисленные в анкете, отличаются от географических и культурных условий Индонезии, например, беговые лыжи, дзюдо, катание на роликовых коньках, горные лыжи и снегоходный спорт.

Целью данного исследования являлась оценка уровня физической активности молодежи. В свете этого описания исследователь хотел бы изучить надежность и валидность опросника Година-Шепхарда "Физическая активность в свободное время".

Материал и методы

В исследовании приняли участие 74 молодых взрослых человека в возрасте 16-25 лет. Для определения конструктной валидности использовался анализ элементов, конвергентная валидность определялась через корреляцию с Глобальным опросником физической активности. Для анализа "тест-ретест" использовалась корреляция с респондентами, повторно заполняющими опросник, а межретестовая надежность определялась через согласованность действий трех рейтеров, которые были определены с помощью специальных критериев.

Результаты

Для анализа данных использовались корреляция Пирсона и коэффициент внутриклассовой корреляции (ICC). Конструктивная валидность показала r>0,5, ретестовая надежность - 0,998, что свидетельствует о валидности и надежности опросника, а ICC - 0,943, что указывает на отличную межретестовую согласованность опросника Година-Шепарда "Физическая активность в свободное время".

Выводы

Согласно результатам исследования, опросник Година-Шепарда "Физическая активность в свободное время" на индонезийском языке является валидным и надежным для использования у лиц молодого возраста.

Ключевые слова: опросник физической активности, валидность, надежность, молодые люди, индонезийская адаптация

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Introductions

Today's society only undertakes a few activities in this modern and practical era; one-third of the global population over 15 is reported to be inadequately active, affecting their health. Several studies show that sedentary behavior in the adult population in developed countries is alarming at 8.3 hours and 7.7 hours for Americans [1,2]. Sedentary lifestyles are becoming a habit almost worldwide due to the lack of space for exercise or sports, an increase in sedentary behavior due to occupations such as working in offices, and Almost all activities in life are dominated by modern instruments due to technological improvements for example increase in utilities that provide videos and television shows. Its impact considerably affects the pattern of human existence and causes disturbances in the behavior of sedentary people. Consequently, there is an increase in health problems in this era [3–6].

Sedentary behavior increases the chance of developing diabetes, hypertension, cardiovascular disease, cancer, and osteoporosis. Sedentary living is a habit of being sedentary or inactive for an extended period with an energy expenditure of less than 1.5 MET. According to research, Americans spend 55% of their active (productive) time sedentary, while Europeans spend 40% of their spare time watching television. [3,7] According to the Ministry of Health of the Republic of Indonesia (2018), the average population of Indonesia with sedentary behavior 6 hours per day is 24.1%, while the population classed as less active in general is 26.2% [8–10].

A measurement tool is required to determine one's degree of physical activity. The Godin-Shephard Leisure-Time Physical Activity Questionnaire estimates the degree of physical activity in spare time [11]. The Godin-Shepard's Leisure Time Physical Activity Questionnaire is an easy-to-understand and practical assessment tool, and it only focuses on the degree of physical activity during leisure time for one week. Physical activity levels are classified into three categories: low, medium, and high [12].

The Godin-Shephard Leisure-Time Physical Activity Questionnaire (Godin-Shepard's Leisure Time Physical Activity Questionnaire) is a practical tool because it has three question points designed exclusively for monitoring leisure time activities. The questionnaire's question points are very appropriate for the character of young adults who enjoy physical activity through hobbies such as basketball, volleyball, and swimming. Additionally, the Godin-Shepard's Leisure Time Physical Activity Questionnaire has been widely used by many researchers in other countries. An instrument's quality cannot be overlooked. Testing the validity of a measurement instrument is crucial because a good instrument can offer optimal trade-offs between quality and feasibility. Using a defective measurement increases the likelihood of misclassification and skewed results in a study [13].

The Godin-Shepard's Leisure Time Physical Activity Questionnaire in Turkish shows strong reliability in evaluating leisure time and Physical activity in diabetic patients aged 40 to 59, with a Cronbach Alpha score of 0.64 indicating that the questionnaire is reliable, and test-retest analysis findings of 0.97 indicating that the questionnaire is reliable. Even when the assessment is repeated, the score shows a genuine result and has a reasonably high consistency value [14].

This measuring instrument, however, is only available in a few languages and has yet to be available in Indonesian. Furthermore, similar questionnaires have yet to be translated into Indonesian, focusing on measuring exercise or sports activity. In addition, some of the physical activity or physical exercise content listed in the questionnaire differs from Indonesia's geographic and cultural conditions, such as cross-country skiing, judo, roller skating, alpine skiing, and snow-mobilizing. Hence, a measuring instrument Godin-Shepard's Leisure Time Physical Activity Questionnaire with high validity and reliability is required. This study aimed to measure the degree of physical activity in young adults. The researcher wants to examine the Godin-Shepard's Leisure Time Physical Activity Questionnaire's reliability and validity in light of this description. The Godin-Shepard's Leisure Time Physical Activity Questionnaire is hoped to help the general public and practitioners in Indonesia understand both the role of a client or a standardized routine measurer and the context of research subjects, especially in physical activity (exercise).

Material and methods

Design and sample

This study sought to determine whether the Indonesian version of the Godin-Shephard Leisure-Time Physical Activity Questionnaire is valid

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and reliable for young adults. The sample in this study comprised of respondents with young adult subjects aged (15-24 years), able to communicate in Indonesian both orally and in writing and willing to participate in various data collection activities. The sample in this study consisted of 74 participants who successfully finished all research series after earlier collecting 81 samples and were willing to participate in this research.

Measures

This study gave respondents forms with data characteristics and the Godin-Shephard Leisure-Time Physical Activity questionnaire for measurement and data collecting.

Form for collecting individual data

The individual characteristics data form includes various questions about the respondents' socio-demographics, such as age, gender, weight, height, level of education, and employment activity. Furthermore, the individual data form contained questions about leisure activities, types of exercise/ exercise/exercise that are habitually conducted, and the respondent's involvement in the sports community.

Godin-Shephard Leisure-Time Physical Activity Questionnaire

It is a measurement tool for assessing a person's physical activity in his leisure time. Compared to other physical activity measurement instruments, the Godin-Shephard Leisure-Time Physical Activity Questionnaire is a measurement tool that is simple to comprehend and practical to use; also, this questionnaire exclusively focuses on the level of physical activity during free time for one week . This instrument was created in 1985. This questionnaire categorizes physical activity levels into mild, moderate, and heavy [15].

The method for calculating the total score in this questionnaire is: Strenuous Activity (9 METs time/week) + Moderate Activity (5 METs time/week) + Light Activity (3 METs time/week). Interpretation of the final score on the Godin Shephard Leisure-Time Physical Activity Questionnaire: A score of more than 24 units is deemed active in free time. A score of 14-23 units is considered moderately active in free time. A score of fewer than 13 units is deemed less active in free time[12]. Several studies demonstrate the reliability of the Godin-Shephard Leisure Time Physical Activity Questionnaire [16].

According to Zelener and Schneider (2016) [17] the validity coefficient of the Godin-Shephard Leisure-Time Physical Activity Questionnaire for 15 days was 0.65 [95% (CI) =.34-.96] and 0.45 for 30 days (95% (CI) =.25-.64). Based on the validity coefficient, the Godin-Shephard Leisure-Time Physical Activity Questionnaire is a valid measurement instrument with a moderate level of validity in evaluating the level of physical activity. The Godin-Shephard Leisure-Time Physical Activity Questionnaire is available in English, Turkish, and Portuguese and has been validated and reliable.

Validity and reliability

Language validity. Forward translation

At this stage, the questionnaire in English was translated by Indonesians who are experts in both English and Indonesian. In the translation process, translators are asked to emphasize concepts rather than literacy, including terms that are not popular in Indonesia. In addition, translations are made in a language that is easy for readers to understand [18,19].

Backward translation

At this stage, the English questionnaire that has been translated into Indonesian is translated back into English by other English language experts without changing the original questionnaire. This stage involves two expert physiotherapists who understand physical activity. Experts are involved in this process to assess the suitability between the activities in the questionnaire and the culture of physical activity in Indonesia; then, the experts ensure that the translation of Indonesian back into English does not change the context of the content according to the original questionnaire. In addition, English and Indonesian language experts also confirmed that the Godin Leisure Time physical activity level questionnaire that had been carried out backward translation did not have a significant difference in meaning, so the questionnaire that had been carried out backward translation had the same meaning as the original questionnaire. The following

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process compares the two versions of the Indonesian questionnaire and the original English questionnaire that has been made and returns the conclusions so that the translation results are obtained in Indonesian, containing items and sentences without changing the meaning and concept based on the original guidelines. [18,20]

The experts researched the suitability of physical activity/exercise contained in the Godin-Shephard Leisure-Time Physical Activity questionnaire, which is commonly used in America and Europe, to fit the sports culture or physical activity in Indonesia, such as hockey, cross-country skiing, judo, rollerblading, baseball, bowling, and snowmobiling. The researchers modified the Godin-Shepard's Leisure Time Physical Activity Questionnaire instrument in numerous ways, including 1) providing the dose by setting the MET value on several exercises, including during severe exercises, such as running 7 Mph, jogging 6 Mph, and cycling 14-16 Mph. Then, with mild exercise, walk at 4 MPH. Then, with light exercise, add a walk of less than 3 MPH. The excluded exercises in hard training or item 1 are hockey, cross-country skiing, judo, and rollerblading. Then, moderate workouts, such as baseball, alpine skiing, popular folk dances, bowling, golf, and snowmobiling, are excluded.

The context of disparities in language, state of health, language, and lifestyle are all crucial factors when adapting to a measurement tool. It is also reported in the literature that it is possible and essential to modify the questionnaire question items based on a region's social and cultural contours. This process will avoid data bias when measuring subjects with different characteristics than the original questionnaire.

Construct validity

Construct validity relates to how successfully the researcher translates or transforms thoughts, ideas, or behaviors that are creations into a reality that performs and operates appropriately [21]. The construct validity form reflects the extent to which construct size variations are theoretically similar and correlated, which is then determined using the Pearson Product Moment coefficient. Three items have been examined. The correlation value analysis (*r* count) can calculate the item-total correlation coefficient with Pearson product-moment. If *r* counts from *the r* table, the instrument (question) is valid [22].

Convergent Validity

Convergent validity is a topic that correlates with construct validity. Convergent Validity can be expressed as the extent to which a measurement instrument is valid when compared with similar instruments widely recognized and with similar constructs. This study employed the Global Physical Activity Questionnaire (GPAQ) for convergent validity. The World Health Organization's Global Physical Activity Questionnaire (GPAQ) is one tool used to assess physical activity levels [23]. The Global Physical Activity Questionnaire was developed to monitor physical activity in developing countries. The GPAQ comprises 16 questions that collect data from research participants in three categories: physical activity at work, travel from one location to another, and recreational or leisure activities[24].

Retest reliability

Test-retest reliability is a retest performed twice with varied execution times on the same instrument in the same group [25]. The consistency of an instrument used to measure anything is referred to as test-retest reliability, and the correlation with each other is calculated using the Pearson product-moment correlation coefficient. The extent of the instrument's reliability coefficient is shown by the correlation of visual scores acquired from the instrument's first and second testing or administration. The researcher's urgency determines the time interval between testing or administering the first and second instruments [26]. The reliability test was repeated twice, with the initial measurement followed by two 24-hour distance measures performed by the same respondent.

Inter-rater reliability

It is used to determine the consistency of raters in assessing an instrument. Three physiotherapists served as raters in this study based on established criteria. The Intraclass Correlation Coefficients (ICC) coefficients are used in the statistical examination of inter-rater dependability. The correlation score can be utilized to determine instrument consistency when performing instrument testing between physiotherapists 1, 2, and 3.

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Data collection

Data for the study were gathered through interviews and the completion of standardized questionnaires on young adult individuals following pre-determined research criteria and programs. During the preliminary research stage, this process will take about 15-20 minutes to complete to provide support and conduct interviews on research topics. In the early stages of the study, a post-content validity process was performed on ten young adult individuals who were not participating in the research process. The team has established a specific space for the interview and application procedure. The data collection and filling process took around four months for 74 respondents, including first filling, test-retest, and inter-rater.

Analysis strategy

IBM SPSS Version 23 was used for statistical analysis. Pearson Product Moment is used to assess construction validity and test-retest reliability. Furthermore, the Intraclass Correlation Coefficients (ICC) were used to examine inter-rater dependability. The inter-evaluator reliability test is a repeat test performed with the same instrument in the same group and at different times with different assessors, in this case physiotherapists.

Ethical clearance

The Godin-Shephard Leisure-Time Physical Activity Questionnaire was used and adapted with permission acquired directly from Godin, the original instrument's developer, via e-mail contact. The study was carried out with the agreement of the Faculty of Medicine's ethics commission. Prior to data collection, the researchers explained the research objectives to the respondents who were used as a sample, and the researchers obtained verbal and written consent from the respondents to be willing to participate.

Results

Table 1 shows the demographic features of research respondents. Based on the total sample, 67.57% of the sample were men, while 32.43% of the sample were women. In the age data, the age range 16-20 years is 64.87%, then the age range 21-25 years is 35.15%, with the average value + standard deviation being 18.45 ± 1,245. Body weight and height data show mean values ± standard deviation, respectively 56.34 ± 11.642 and 129.46 ± 60.399. Body mass index data show a percentage of 18.91%, with an average body mass index of 64.86%. Furthermore, the body mass index with overweight was 14.86%, and the body mass index with obesity classification was 1.34%.

Table 2 explains the results of the statistical analysis of the validity and reliability of the Godin Shephard Leisure Time Physical Activity Questionnaire. In the convergent validity test with the comparator, the Global Physical Activity Questionnaire, the value r=0.775 (r table=0.1927) means the questionnaire is valid with a strong correlation. Furthermore, the results of the statistical validity analysis of the construct with analysis of items one, two, and three show respectively the value of r=0.787; r=0.746; r=0.598 shows the questionnaire is valid with strong and quite strong correlation. The interrater reliability aspect (test-retest) shows a value of r=0.998, meaning it is reliable with a powerful relationship. The interrater reliability test shows a p-value of 0.735 and an ICC of 0.943, which means that the Godin Shephard Leisure Time Physical Activity Questionnaire is consistent.

Table 1
Descriptive Statistics of Survey Indicators among Indonesian respondents.

Characteristics	N	%	Mean ± STDEV
Sex	74		
Male	24	32.43%	
Female	50	67.57%	
Age	74		1.245
16-20 years old	48	64.87%	
21-25 years old	26	35.13%	

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Characteristics	N	%	Mean ± STDEV
Weight (kg)	74		56.34 ± 11.642
Height (cm)	74		129.46 ± 60.399
Body Mass Index (BMI) (kg/m^2)	74		
Underweight	14	18.91%	
Normal	48	64.86%	
Overweight	11	14.86%	
Obesity	1	1.34%	
Godin-Shepard's Leisure time physical activity questionnaire (Units)	74		
First test			32.92 ± 17.487
Second test			33.41 ± 17.945
Godin-Shepard's Leisure time physical activity questionnaire interpretation (Units)	74		
Active	50[a]	67.57%[a]	
	52[b]	70.27%[b]	
Moderate	20[a]	27.03%[a]	
	19[b]	25.68%[b]	
Sedentary	4[a]	5.40%[a]	
	3[b]	4.05%[b]	

Notes: a – first test; b – second test

Table 2

The results of validity and reliability test of The Godin Shephard Leisure Time Physical Activity Questionnaire

Indicators	*p* value	*r*	*r* table	A	ANOVA	Cronbach's Alpha	ICC	Conclusion
Normality Distribution								
Question 1	0.082							Normal
Question 2	0.352							Normal
Question 3	0.276							Normal
Total Godin-Shepard's Leisure time physical activity questionnaire (Units)	0.174							Normal
1st Test	0.174							Normal
2nd Test	0.188							Normal
1st Rater	0.235							Normal
2nd Rater	0.328							Normal
3rd Rater	0.254							Normal
Global Physical Activity Questionnaire (METS)	0.148							Normal
Convergent Validity Godin-Shepard's Leisure time physical activity questionnaire (Units) vs Global Physical Activity Questionnaire (METS)		0.775	0.1927	0.05				Valid
Construct validity								
Question 1		0.787	0.1927	0.05				Valid
Question 2		0.746	0.1927	0,05				Valid
Question 3		0.598	0.1927	0.05				Valid
Retest reliability Godin-Shepard's Leisure time physical activity questionnaire (Units)		0.998	0.1927	0.05				Reliable
Inter-rater reliability Godin-Shepard's Leisure time physical activity questionnaire (Units)	0.735				0.735	0.980	0.943	Consistent

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Discussion

This study examines the validity, reliability of test re-test, and reliability among raters of the Godin-Shephard Leisure-Time Physical Activity Questionnaire of 74 young adult respondents. The researchers employed a self-questionnaire system in this investigation. Filling out a questionnaire independently based on one's judgment and competence is called self-questionnaire [27]. A self-questionnaire can prompt respondents to characterize their tendencies based on their feelings. In terms of the type of exercise being carried out, many factors must be modified to the situation in Indonesia due to changes in questionnaire interpretation based on current cultural contexts, which alter the results provided by the measuring device [28,29].

The validity of the Godin-Shepard's Leisure Time Physical Activity Questionnaire was tested using Pearson product moment analysis for each item question or question item on the overall score; the results show that the three items on the Godin-Shepard's Leisure Time Physical Activity Questionnaire are deemed valid because the r count is larger than the value. Item 1 is in the heavy exercise category with a value of 0.832, indicating a powerful validity interpretation; item 2 is in the Moderate Exercise category with a value of 0.830, indicating a powerful validity interpretation; and item 3 is in the mild exercise category with a value of 0.840, indicating a powerful validity interpretation [30].

Other than measurement instruments, elements influencing the validity of instruments include the measuring instrument users who do the measurements and the people being measured [31]. Researchers utilized adolescent volunteers in this study, which aided researchers in filling out the questionnaire because the respondent's degree of knowledge could be deemed either one or the other. The cause was observed when filling out the questionnaire went well, and several participants actively asked questions when they felt they did not comprehend specific areas in the Godin-Shepard's Leisure Time Physical Activity Questionnaire.

According to the study findings, 5.40% of participants reported an "inactive" level of physical activity, sometimes a sedentary lifestyle. Respondents in this group are all overweight or obese. These findings are consistent with those of [32], who discovered that a sedentary or passive lifestyle is an independent determinant of weight increase. Sedentary activity is frequently associated with watching TV, playing smartphone games, or being too lazy to walk. According to research, regular physical activity and proper weight control can minimize the risk of heart disease and premature death while also improving general physical fitness [33]

Young individuals need a higher understanding of the importance of physical activity in particular dosages [8]. Gaston Godin created the Godin-Shephard Leisure-Time Physical Activity Questionnaire. Several items in the Godin-Shepard's Leisure Time Physical Activity Questionnaire are managed to find information about how many times a person is involved in light (minimal effort), moderate (not tiring), and heavy (fast heart beating) LTPA in a minimum of 15 minutes over seven days [12].

The study's startling discovery was that those in the active and moderately active categories at the level of physical activity participation revealed data of 67.57% and 27.03%, respectively. In comparison, individuals in the inactive participation rate were 5.4% [34]. According to the findings of the study, respondents were more active in physical activities as a result of the COVID-19 pandemic in the previous year, which motivated to boost the body's immune system, vaccine policies, and 'new normal' government policies to relax outdoor community activities and isolation boredom inside the house [35].

After testing the convergent validity of the Global Physical Activity Questionnaire, it was found that Godin-Shepard's Leisure time physical activity questionnaire was valid. This questionnaire assesses all three physical activity domains: occupational activity domains, travel to and from certain places, and recreational activities. The similarity of this questionnaire with The Global Physical Activity Questionnaire is that it assesses the level of physical activity, divided into light, moderate, and heavy. However, Godin-Shepard's Leisure time physical activity questionnaire focuses on assessing recreational activities. The Godin-Shepard's Leisure time physical activity questionnaire has also been tested for convergent validity with the correlation of physical fitness indicators, namely VO2 max and Body Mass Index. Of course, Global Physical Activity Questionnaire and Godin-Shepard's Leisure time physical activity questionnaire have been tested with the Metabolic Equivalents score [36].

These incentives motivate the community,

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particularly young individuals, to resume outdoor activities, particularly exercise, to enhance health and physical fitness. Several studies suggest that physical activity effectively responds to immune system regulation against infectious illnesses [6]. The community's trend of restarting outdoor sporting activities with the community stimulates the promotion of physical activity, demonstrating that the environment influences individual motivation to conduct physical activity [37].

Based on the re-test reliability test findings, which revealed strong reliability (r=0.998), this questionnaire adaption demonstrates that this questionnaire can consistently evaluate respondents' level of physical activity (particularly exercise activity in spare time). This study's participants were young individuals aged 18 to 25. Respondents were generally unfamiliar with measuring and evaluating physical activity levels, making it difficult to understand and interpret each question item during the research, necessitating additional descriptions for several respondents to ensure they could answer correctly according to the actual report.

In this case, the respondent's difficulty was understanding the level of physical activity based on an appropriate description based on the respondent's perception, even though cultural studies had previously been conducted to adjust the types of physical activity that were not suitable in Indonesia, such as skiing, judo, bowling, and snowmobiling.

The Godin-Shepard's Leisure Time Physical Activity Questionnaire is a questionnaire used to assess physical activity levels over one week [38]. As a result, various sports have been altered to fit the culture of Indonesia. According to the literature, culturally changing an instrument is indeed vital to safeguard the equity of each language and culture [39].Physical activity features are adjusted through cultural adaptation in the form of sports often practiced in Indonesia, such as football, badminton, futsal, jogging, cycling, and volleyball [40]. Adaptation modifications will aid comprehension and adjustment in filling out the questionnaire, as well as determining whether the translation process of the original version of the questionnaire with the Indonesian version has stayed the same meaning and concept.

The interrater reliability test yielded a *p-value* of 0.735, a Cronbach alpha coefficient of 0.980, and an Interclass Correlation Coefficient (ICC=0,943), indicating that this questionnaire has high consistency when evaluated by various raters. Three physiotherapists used preset criteria to perform the inter-rater reliability test in this study. The data from the interrater test demonstrated that the Godin-Shepard's Leisure Time Physical Activity Questionnaire questionnaire was consistent between assessors.

Even though the test results demonstrated good/reliable consistency, there was a gap in the results of completing the questionnaire accompanied by three raters, as seen in the standard deviation results; the first rater was STDEV 18.616, the second rater was STDEV 20.946, and the third physiotherapist was STDEV 21.013. This variance is influenced by the three raters' varying explanations of the Godin-Shepard's Leisure Time Physical Activity Questionnaire questionnaire and the respondent's ability to create consistency in recalling the physical activities completed in the preceding week. The findings of this study are congruent with those of [14],who investigated the Turkish adaption of the Godin Leisure-Time Exercise Questionnaire. The results of the interrater reliability test indicate that this questionnaire has high consistency when assessed by different raters (ICC = 0.98).

People's exercise frequency varies according to culture and habit. Previous research has shown that a metropolitan environment with rapid economic growth and widespread use of modern technology influences people's participation in physical activity/ sports, particularly the tendency to be sedentary or inactive, as demonstrated a study which found that physical activity in the United States is generally lower than in Finland, Germany, and Sweden. However, this declining trend is not limited to developed nations [41,42]. People in underdeveloped countries are getting less physically active, according to research. Although most individuals in developing nations continue to work and engage in travel activities that involve physical activity in their daily lives, increasing automation in work and personal life is essential in engaging in physical exercise [43,44].

The lack of physical activity alternatives in the Godin-Shepard's Leisure Time Physical Activity Questionnaire questionnaire caused respondents to feel confused when filling it out. However, this was alleviated by the presence of a rater when filling out the questionnaire. Another constraint is a rater's lack of experience; hence, choosing a rater with more advanced requirements, expertise, and good experience is preferable. More raters are looked for

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in future studies since the more raters used, the better or more accurate the interrater reliability coefficient will be [45].

Conclusions

The Godin-Shephard Leisure-Time Physical Activity Questionnaire has been adapted to Indonesian culture, including a study on the applicability of terminology and language in physical activity and the validity of the examined material. The adaption of this questionnaire has been examined by professionals and is tailored to the regular physical activities and exercises commonly performed. During the process, several experts' perspectives on the qualities of physical exercise suited in Indonesia differed. Adaptation questionnaires in Indonesian were shown to be valid, trustworthy, and simple to administer to young adult individuals. Researchers urge that this study be expanded with a more significant number of various sample groups to assess the applicability of this questionnaire to different populations in society.

Conflict of interest

The authors declare that there is no conflict of interest.

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Information about the authors

Farid Rahman
fr280@ums.ac.id
https://orcid.org/0000-0001-7397-4306
Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta 57162, Indonesia

Brissa Isabel Salsabila
j120221092@student.ums.ac.id
https://orcid.org/0009-0007-5644-7089
Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta 57162, Indonesia

Ilham Setya Budi
ilhamsetya37@gmail.com
https://orcid.org/0000-0003-1548-6034
Dr. Muh. Sani Hospital, Kepulauan Riau, Karimun 29661, Indonesia

Dessy Triana Ramadhani
j120180023@student.ums.ac.id
https://orcid.org/0000-0002-9616-0425
Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta 57162, Indonesia

Syarif Hidayat
j120170023@student.ums.ac.id
https://orcid.org/0009-0002-1487-940X
Department of Physiotherapy, Faculty of Health Science, Universitas Muhammadiyah Surakarta, Surakarta 57162, Indonesia

Azliyana Azizan
azliyana9338@uitm.edu.my
https://orcid.org/0000-0002-0548-3975
Department of Physiotherapy, Faculty of Health Science, Universiti Teknologi Mara Shah Alam, Selangor 40450, Malaysia

Інформація про авторів

Фарид Рахман
fr280@ums.ac.id
https://orcid.org/0000-0001-7397-4306
Кафедра фізіотерапії, Факультет наук про здоров'я, Університет Мухаммадія Сураката, Сураката 57162, Індонезія

Брісса Ізабель Сальсабіла
j120221092@student.ums.ac.id
https://orcid.org/0009-0007-5644-7089
Кафедра фізіотерапії, Факультет наук про здоров'я, Університет Мухаммадія Сураката, Сураката 57162, Індонезія

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Ільхам Сетья Буді
ilhamsetya37@gmail.com
https://orcid.org/0000-0003-1548-6034
Доктор Мух. Лікарня Сані, Кепулауан Ріау, Карімун 29661, Індонезія

Дессі Тріана Рамадхані
j120180023@student.ums.ac.id
https://orcid.org/0000-0002-9616-0425
Кафедра фізіотерапії, Факультет наук про здоров'я, Університет Мухаммадія Сураката, Сураката 57162, Індонезія

Сіаріф Хідаят
j120170023@student.ums.ac.id
https://orcid.org/0009-0002-1487-940X
Кафедра фізіотерапії, Факультет наук про здоров'я, Університет Мухаммадія Сураката, Сураката 57162, Індонезія

Азліяна Азізан
azliyana9338@uitm.edu.my
https://orcid.org/0000-0002-0548-3975
Кафедра фізіотерапії, Факультет наук про здоров'я, Технологічний університет Мара Шах Алам, Селангор 40450, Малайзія

Информация об авторах

Фарид Рахман
fr280@ums.ac.id
https://orcid.org/0000-0001-7397-4306
Кафедра физиотерапии, факультет здравоохранения, Университет Мухаммадия Сураката, Сураката 57162, Индонезия

Брисса Изабель Сальсабила
j120221092@student.ums.ac.id
https://orcid.org/0009-0007-5644-7089
Кафедра физиотерапии, факультет наук о здоровье, Университет Мухаммадия Сураката, Сураката 57162, Индонезия

Ильхам Сетя Буди
ilhamsetya37@gmail.com
https://orcid.org/0000-0003-1548-6034
Доктор Мух. Больница Сани, Кепулауан Риау, Каримун 29661, Индонезия

Десси Триана Рамадхани
j120180023@student.ums.ac.id
https://orcid.org/0000-0002-9616-0425
Кафедра физиотерапии, факультет здравоохранения, Университет Мухаммадия Сураката, Сураката 57162, Индонезия

Сиариф Хидаят
j120170023@student.ums.ac.id
https://orcid.org/0009-0002-1487-940X
Кафедра физиотерапии, факультет здравоохранения, Университет Мухаммадия Сураката, Сураката 57162, Индонезия

Азлияна Азизан
azliyana9338@uitm.edu.my
https://orcid.org/0000-0002-0548-3975
Кафедра физиотерапии, факультет медицинских наук, Технологический университет Мара Шах Алам, Селангор 40450, Малайзия

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